

LEGO SPIKE Prime Robots

WHAT'S IN THE KIT

- 4 LEGO SPIKE Prime kits — hub, motors, color/force/distance sensors, technic elements



QUICK-START GUIDE

1. Open the SPIKE app (Chromebook-friendly) and pick a unit.
2. Guided builds pair step-by-step instructions with coding lessons.
3. Blocks first; Python is available for advanced students.
4. Sort-by-tray cleanup keeps 500+ pieces sane.

FULL LESSONS IN THIS GUIDE

GRADES 3-5	Helpful Inventions	page 2
GRADES 6-8	Sensor Olympics	page 3

PRO TIP: 4 kits = ideal for a station or club; pair with VEX IQ for a whole-class robotics unit.

LEGO SPIKE Prime Robots

LESSON · GRADES 3-5

Helpful Inventions

LEARNING OUTCOMES

- Students will build and program a functional robot from guided plans
- Students will explain how their machine helps a user
- Students will propose an evidence-based improvement

MATERIALS

- LEGO SPIKE Prime kits (4 stations)
- Devices with the SPIKE app
- Guided unit selection (sorting machine, delivery cart, helper bots)
- Improvement proposal sheet

INSTRUCTIONS

1. Stations run a guided unit build with the app's step-by-step instructions.
2. Program the guided behaviors, then test with real objects.
3. Frame the empathy question: who does this machine help, and how?
4. Teams demo their helper and answer audience questions.
5. Improvement proposals: one concrete change, with the reason it would help the user more.
6. If time allows, implement the top-voted improvement.

ACTIVITY

Helpful Inventions — build a helper robot, then defend how it helps and how it could help better. Engineering with a user in mind, from the very first build.

LEGO SPIKE Prime Robots

LESSON · GRADES 6-8

Sensor Olympics

LEARNING OUTCOMES

- Students will program with sensor input for precise control
- Students will isolate variables when tuning robot behavior
- Students will compete on measured accuracy across events

MATERIALS

- LEGO SPIKE Prime kits
- Devices with SPIKE app (blocks; Python for advanced)
- Event courses: line to follow, distance targets, object detection zone
- Event scoresheets

INSTRUCTIONS

1. Present the three Olympic events: line following, precise-distance driving, object detection stop.
2. Each event isolates one sensor: color, motor rotation, distance.
3. Teams train event by event, tuning one parameter at a time.
4. Qualifying runs establish baseline scores; tuning continues.
5. Olympics day: three official scored attempts per event.
6. Medal debrief: which tuning strategies transferred across events?

ACTIVITY

Sensor Olympics — three events, three sensors, scored on precision. Isolating one sensor per event builds real control intuition before students ever combine them.